



Emergence Of GaAs Devices As A Substitute To Silicon

Less power consumption and more efficiency offered by GaAs wafers are attracting the market players to adopt them, thereby increasing the demand for GaAs wafers. Previously, the optoelectronic devices were broadly used in short-range optical communications and computer peripherals but now these are in demand for some emerging applications such as lidar, augmented reality, and face recognition as well

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The recent global shortage of semiconductor chips is playing havoc with the production of electronics goods including white goods and automobiles, which now use lots of these components. The demand has surged far beyond production, which got reduced due to lockdowns and other issues due to Covid-19 all over the world.

With the increasing demand and low supply of electronic basic material such as silicon wafers, scientists, engineers, and manufacturers dealing with electronics industry are always on the lookout for substitute material. Gallium in its combinations as gallium nitride (GaN) and gallium arsenide (GaAs) has emerged as a strong candidate, even if not a replacement for silicon, to at

least supplement its shortage with more diverse applications in electronics industry.

Energy savings have become the buzz word in every sphere of our life due to our heavy dependence on energy, its bad environmental effects as well as the depleting resources. Electronic devices as an integral part of our present-day lifestyle can also be a major source of energy savings. But a significant amount of electric energy is being lost during power electronics processes, thus limiting both power efficiency and reliability of electrical and mechanical systems.

Currently, the three semiconductor materials being used in power electronics are silicon (Si), silicon carbide (SiC), and gallium nitride (GaN). Among these, silicon does not perform well in higher energy density and temperature conditions, and SiC and GaN cannot be used for higher currents and are often too expensive. Therefore, gallium arsenide (GaAs) (see Fig. 1) is widely used in microelectronics due to its suitability for high frequency and temperature, but not for controlling and converting electrical energy.

Science of GaAs

GaAs is a group III-V direct band gap semiconductor having a zinc blende crystal structure. Since GaAs is a compound, each gallium atom in the structure is surrounded by arsenic atoms, and similarly gallium atoms surround each arsenic atom in the structure as shown in Fig. 2 of unit cell structure. Three valence electrons of gallium atoms share with five valence electrons of arsenic atoms. In this way, each of the GaAs atoms gets eight electrons in its outermost shell.

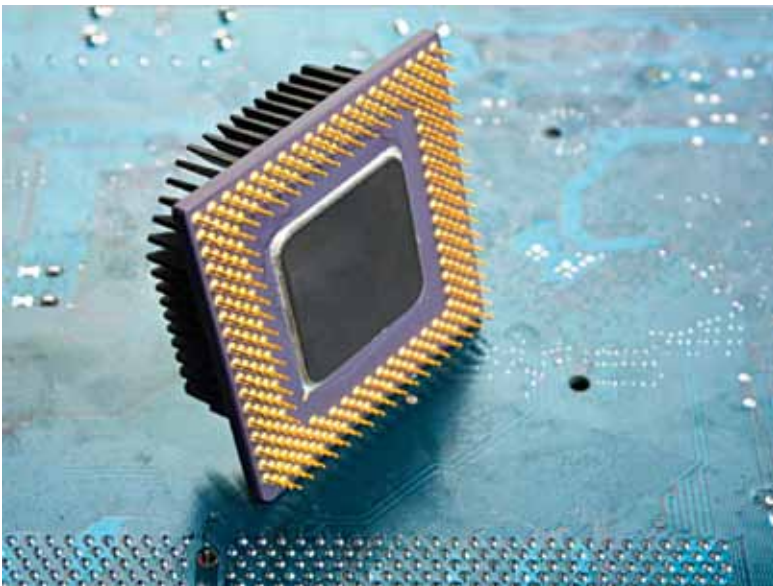


Fig. 1: GaAs wafer (Credit: www.waferworld.com)